Tectonophysics, 184 (1990) 137–156
Elsevier Science Publishers B.V., Amsterdam

The uplift magnetisation record in the central Scandinavian Caledonides: results from basic igneous complexes

J.D.A. Piper [a], T.J. Poppleton [a], R. Mason [b] and A.J. Griffiths [c]

[a] *Geomagnetism Laboratory, Department of Earth Sciences, University of Liverpool, P.O. Box 147, Liverpool L69 3BX (United Kingdom)*
[b] *Department of Geological Sciences, University College, Gower Street, London WC1E 6BT (United Kingdom)*
[c] *Marconi Underwater Systems Ltd., Blackmoor Lane, Watford, Hertfordshire WD1 8YR (United Kingdom)*

(Received November 9, 1989; revised version accepted April 26, 1990)

ABSTRACT

Piper, J.D.A., Poppleton, T.J., Mason, R. and Griffith, A.J., 1990. The uplift magnetisation record in the central Scandinavian Caledonides: results from basic igneous complexes. *Tectonophysics*, 184: 137–156.

To identify the palaeomagnetic record acquired during slow post-orogenic uplift and cooling in the central part of the Scandinavian Caledonides we have investigated four separated synorogenic basic complexes in the Middle and Upper Allochthons. A shallow N–S axis of magnetisation is common to three of these bodies and therefore postdates all major tectonism. It is defined in the Møsjoen Gabbro ($D/I = 7°/3°$, $N = 23$ samples, $\alpha_{95} = 5.0°$), in the Sulitjelma Gabbro ($D/I = 20°/-9°$, $N = 22$ samples, $\alpha_{95} = 6.6°$) and the Råna norite ($D/I = 9°/-16°$, $N = 19$ samples, $\alpha_{95} = 7.9°$). No vestige of this uplift remanence is recognised in the Lyngen Ophiolite, where a characteristic remanence of $D/I = 65°/43°$ ($N = 19$ samples, $\alpha_{95} = 10.6°$) is recorded. The possibility that the N–S shallow axis of magnetisation is of mid-Carboniferous age, is considered and rejected. Instead it is attributed to the interval of post-tectonic cooling of the Caledonian nappe pile between synorogenic events dated ca.. 420 Ma and the uplift to surface levels defined by later Devonian events. It corresponds closely to an axis of magnetisation extensively represented within the metamorphic Caledonides and Scotland and believed to record a post-tectonic thermochemical event. Subordinate components record Mesozoic overprinting events and relict Caledonian magnetisations which have escaped replacement during this thermo-tectonic event. The latter collectively imply a rapid movement from high to low palaeolatitudes in Late Ordovician and Early Silurian times.

Introduction

The emerging recognition of the importance of magnetic overprinting in Palaeozoic rocks during Carboniferous and Permian times (e.g., Creer, 1968; Irving and Strong, 1985; Kent and Opdyke, 1985) has forced a radical reassessment of the palaeomagnetic record from these times and questioned palaeogeographic models developed from the pre-late Carboniferous record (Piper 1987). In particular, it has highlighted the need to isolate areas and rock facies which have preserved a Palaeozoic palaeomagnetic record predating Permo-Carboniferous remagnetisation events. Situations in which remagnetisation has been profound are important in their own right because they lead to recognition of regional tectonic, thermal and diagenetic events that may not be apparent from the application of other methodologies (e.g., Miller and Kent, 1986).

Of special interest in the former context is the record from the Caledonides of Northwest Europe; it has been evident for more than two decades that a substantial part of the palaeomagnetic imprint in the terranes comprising this orogen contrasts with the record from "cratonic" Europe, and has largely survived overprinting in later palaeofields (Briden et al., 1973).

Within this sector a Siluro-Devonian direction first isolated by Chamalaun and Creer (1964) and characterised by NE^-/SW^+ directions has since been found by numerous studies of Upper Silurian–Lower Devonian volcanics and sediments in Britain (e.g., Sallomy and Piper, 1973a;

0040-1951/90/$03.50 © 1990 – Elsevier Science Publishers B.V.

McClelland Brown, 1983; Torsvik, 1985a) and substantiated by extensive field tests. An important complementary body of data has been derived from the metamorphic Caledonides in Scotland which yield a record acquired in a slowly-cooled environment during post-tectonic uplift (Watts, 1982; Torsvik et al., 1983; Esang and Piper, 1984; Watts and Briden, 1984; Watts, 1985). These latter results provide both a long temporal record of the migration of the palaeofield vector and a critical comparison with magnetisations acquired in contrasting conditions by supracrustal rocks. Although it is typically difficult to define the age and duration of remanence acquisition during slow post-tectonic uplift and cooling, these studies identify the importance of a N–S shallow of magnetisation which appears to record a widespread late Caledonian thermochemical event (Esang and Piper, 1984; Torsvik, 1985b).

Only a few palaeomagnetic studies have yet been made of the lateral continuation of the metamorphic Caledonides in Scandinavia. They include a study of the amphibolitised margin of the Sulitjelma Gabbro (Piper, 1974), an investigation of the Fongen–Hyllingen Gabbro (included in the Meraker nappe of the Upper Allochthon, Fig. 1)) by Abrahamsen et al. (1979) and a study of metadolerites in the Sarv nappe of western Sweden by Bylund and Zellman (1980). Further progress in this sector of the orogen is unlikely to be achieved by study of single geological units because only with a distributed sample can the record acquired during post-orogenic uplift and cooling be recognised: since peak metamorphic temperatures were typically somewhat below the highest parts of the blocking temperature spectra of the magnetic carriers, it is possible that some remanence components have survived the metamorphism of the nappe pile and record tectonic displacements during emplacement of the nappes in the culminating (Scandian: Silurian–Early Devonian, ca. 420–400 Ma) phases of orogenic deformation. This possibility was suggested by some results of the studies by Abrahamsen et al. (1979) and Bylund and Zellman (1980) and has been further discussed by Abrahamsen (1985). Since local field control cannot generally isolate tectonically-influenced magnetisation within individual bodies, in this study we have investigated four separated basic units to isolate that common part of the remanence record in the orogen acquired during the post-orogenic cooling.

Geological setting

The Scandinavian Caledonides comprise a metamorphosed nappe pile stacked by thrust movements directed NW–SE onto the Fenno-scandian (Baltic) Precambrian platform (Fig. 1) in an orogenic event culminating in Middle–Upper Silurian times (Roberts and Gee, 1985). The nappes are grouped into four major complexes which comprise the outcrop from east to west of the Lower, Middle, Upper and Uppermost Alloch-

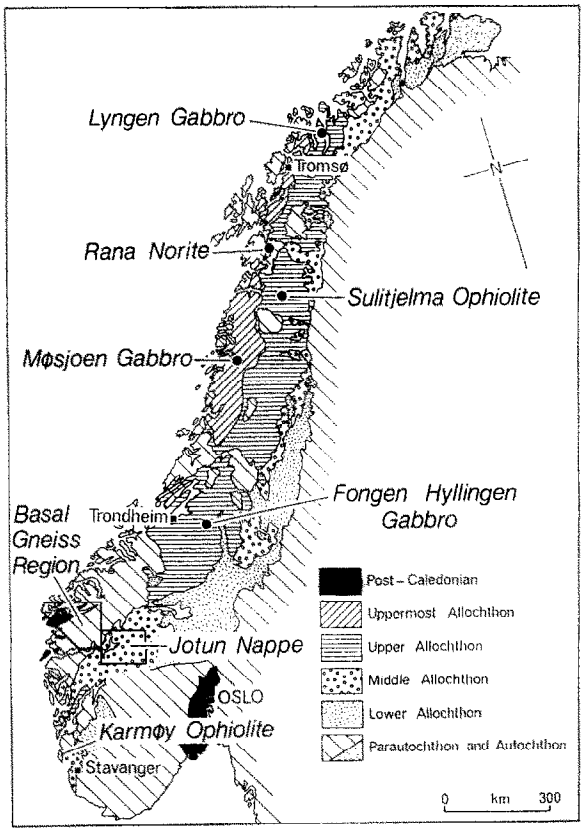


Fig. 1. Simplified tectonostratigraphic map of the Scandinavian Caledonides after Roberts and Gee (1985) showing the locations of the basic units discussed in this paper. The Basal Gneiss Region (BGR) and Jotun nappe in the south central Caledonides form the subject of a comparative palaeomagnetic study (Piper and Poppleton, 1990) discussed in the regional assessment section.

thons. The Lower Allochthon comprises a wedge of Late Precambrian and Early Palaeozoic sediments detached and translated on a basal sole thrust traceable along the entire length of the orogen. The Middle Allochthon comprises predominantly heavily mylonitised slices of Precambrian basement metamorphosed to middle and upper greenschist facies; these rocks record much higher strains than the Lower Allochthon. The Upper Allochthon contains extensive evidence of igneous activity with successions derived from a variety of destructive plate margin environments linked to the former site of the Iapetus Ocean (Hossack and Cooper, 1986); it includes the Sulitjelma and Lyngen ophiolites and the Råna Norite of this study. These rocks record deformations linked to Early Ordovician (Finnmarkian) and Siluro-Devonian (Scandian) tectonic episodes and the metamorphic grade reaches upper amphibolite facies or higher. The Uppermost Allochthon comprises nappes incorporating both metamorphic basement and a range of supracrustal rocks. A variety of synorogenic granites and the Møsjoen Gabbro of this study have been emplaced into these nappes. Nappe emplacement was succeeded by uplift, erosion and deposition in Early and Middle Devonian times of molasse deposits in fault controlled basins; there were, in turn, influenced by an episode of folding, thrusting and low grade metamorphism (the Solundian episode) in late Devonian to Early Carboniferous times.

Palaeomagnetic results

Sampling and laboratory procedures

Palaeomagnetic sites are illustrated in Fig. 2; they typically comprise between seven and twelve independently-oriented cores collected with a portable motorised drill and oriented by sights on the Sun together with magnetic bearings in conjunction with topographic sights. Cores sliced into 2.4 cm-long cylinders in the laboratory were measured with parastatic or cryogenic magnetometers before and after progressive a.f. or thermal demagnetisation. Components of magnetisation are recognised by analysis of vector plots following initial study of laboratory stereoplots; only vectors defined by three or more steps of progressive treatment are analysed here.

The Møsjoen Gabbro

The Møsjoen Gabbro is a synorogenic body emplaced into the Helgeland nappe of the Uppermost Allochthon south west of Mo i Rana at 13.3° E, 65.9° N (Fig. 1, and Stephens et al., 1985b). It has an approximately ovoidal outcrop ca. 10 km × 15 km in size and cuts marble and schist members of a cover sequence metamorphosed to amphibolite facies. Migmatites are developed at the eastern margin and late tectonic granitic dykes emplaced into the nappe yield Rb–Sr ages of 420 Ma (data of Trondbakken quoted in Stephens et al., 1985a).

The palaeomagnetic sample comprises 12 sites on an E–W traverse spanning the width of the intrusion with the majority collected along the margins of Fustvatn (Fig. 2a, Table 1). Two sites in fine-grained amphibolitised units (1, 6) yielded no significant remanence vectors. Sites 2–5 from gabbro facies in the western margin of the pluton defined consistent northerly shallow ("A") directions held by moderately strong remanence components ($15–0.2 \times 10^{-5}$ A m^2 kg^{-1}), progressively subtracted by a.f. and thermal cleaning (Fig. 3). These components have uniformly high coercivities (J_{20}/J_0, the ratio of remanent intensity after cleaning in 20 mT and the initial remanence, = 0.9–0.3), but are not consistently recovered above the Curie point of magnetite and are evidently resident in this phase; this is confirmed by thermal demagnetisation (Fig. 3) and further suggested by IRM and thermomagnetic studies. This direction of magnetisation is not represented in the eastern part of the gabbro (sites 8–12) where an E-positive direction is held by components which typically somewhat stronger ($13–5 \times 10^{-5}$ A m^2 kg^{-1}) than the western sites. These components are of high stability, and subtracted by a.f. cleaning but have lower coercivities (J_{20}/J_0 = 0.2–0.01) than the "A" group; this remanence is unblocked by thermal cleaning below 590° C (Fig. 3). The population of directions does not approximate to a circular distribution (Fig. 5): although discrete convergent components are de-

fined by cleaning and their inclinations are ca. +50°, the declinations range from SE to NE. The example from site 7 in Fig. 3 shows both directions with the E-directed component in lower coercivity fraction. Three of these sites (10–12) occur within 150 m of each other and a local tectonic explanation cannot apparently explain this distribution of declinations.

Three representative mean directions of magnetisation are listed in Table 2; they employ sample means on the assumption that these components were acquired over a protracted period of time during uplift and cooling and therefore record time-averages of the ancient field rather than virtual geomagnetic directions. The "A" component represents the only component recorded by

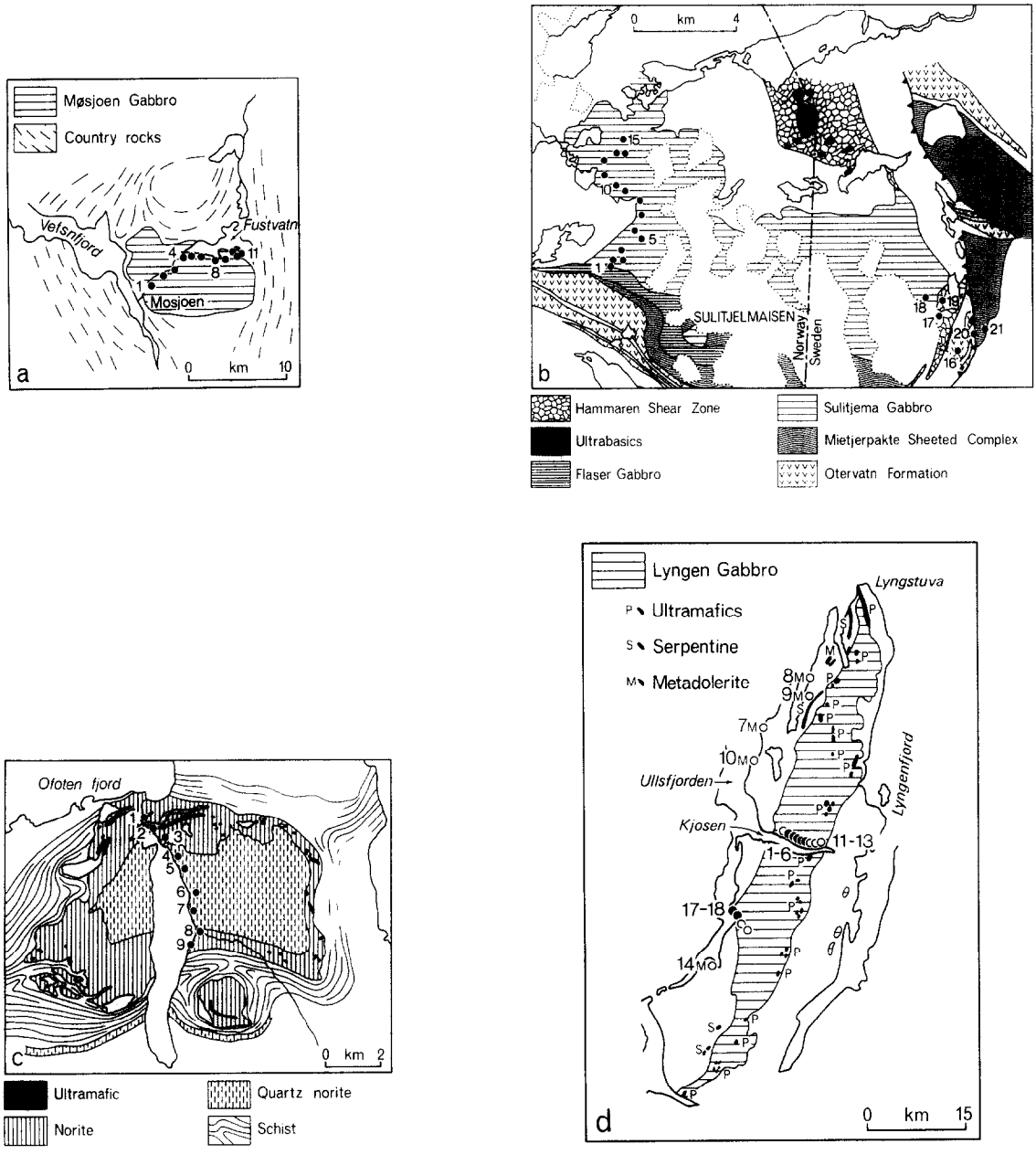


Fig. 2. Simplified geological maps (with the locations of palaeomagnetic sampling sites) of (a) the Møsjoen Gabbro, (b) the Sulitjelma Gabbro, (c) the Råna Norite and (d) the Lyngen Ophiolite.

the western sites (2–5) and has a mean of $D = 7°$, $I = 3°$ (23 samples, $\alpha_{95} = 5°$). The "B" component is an easterly positive group recorded mostly at sites 9–12 ($D = 74°$, $I = 49°$) and "C" is a minor group found mostly at sites 8, 11 and 12 but comprising highly stable samples with the same inclination but a southeasterly declination ($D = 140$, $I = 51°$). There is no clear evidence for a smearing between the "B" and "C" groups (Fig. 5). All components resolved (following cleaning)

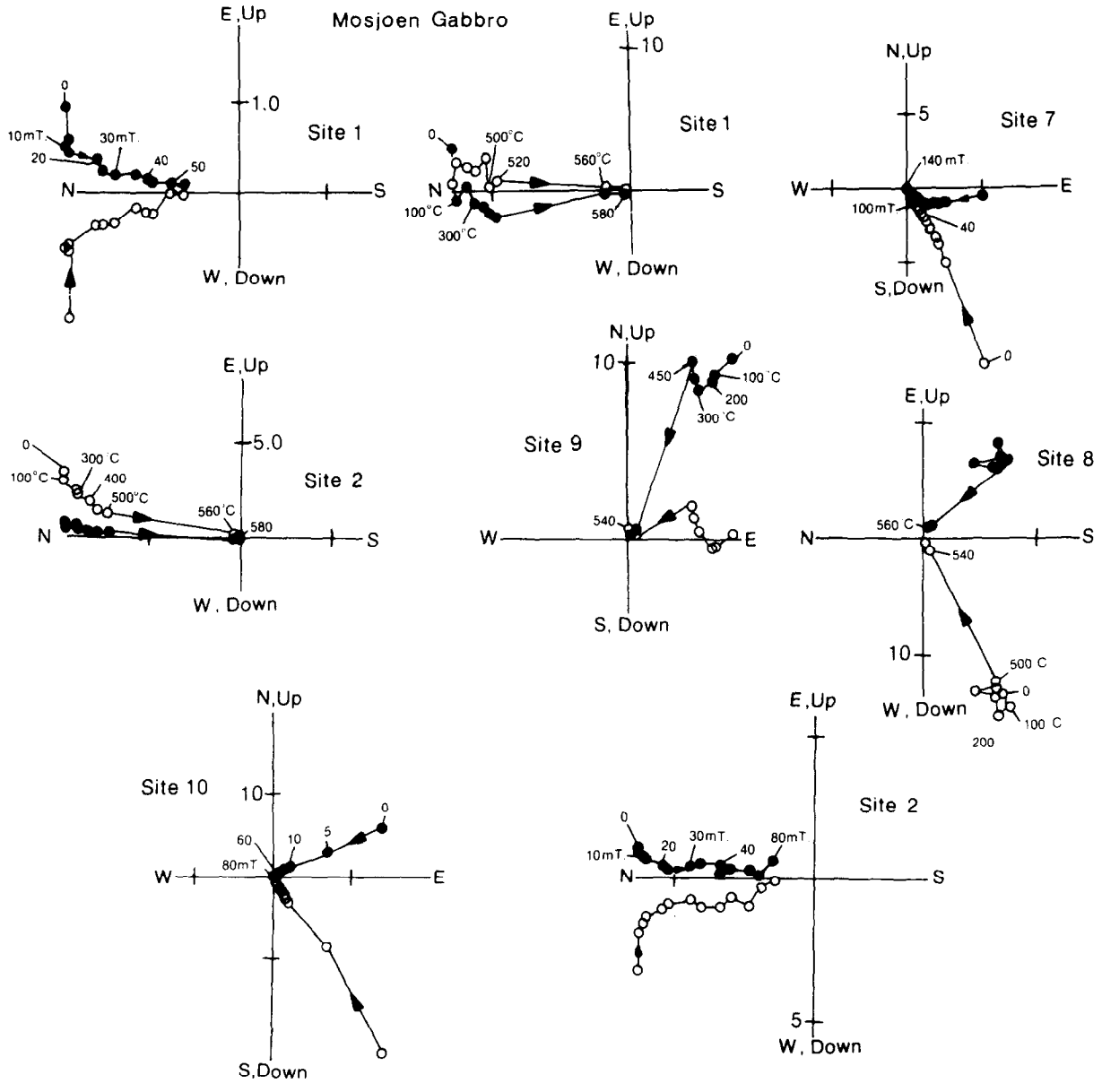


Fig. 3. Representative components of magnetisation resolved from the Møsjoen Gabbro. Sites 1 and 2 illustrate examples of shallow northerly components isolated by a.f. and thermal cleaning. The sample from site 7 illustrates an easterly positive component subtracted by a.f. cleaning to recover a SE component with similar inclination also subtracted by a.f. cleaning; site 9 is an example of thermal cleaning of the former component and site 8 of the latter component. Site 10 is an example of a discrete component with the same inclination and a NNE declination. The solid symbols refer to projections onto the horizontal plane and the open symbols to projections onto the vertical plane; the units of intensity on the axes refer to 10^{-5} A m^2 kg^{-1}. The site numbers refer to locations shown in Fig. 2 and discussed in the text.

TABLE 1

Site mean directions, Caledonian basic complexes, Norwegian Caledonides *

Site No.	Remanence directions					
	polarity	N	D (°)	I (°)	k	α_{95} (°)
Møsjoen Gabbro						
2	N	7	7	−2	84	7
3	N	6	4	0	54	9
4	N	4	12	25	48	13
5	N	6	4	0	102	7
7	N	5	35	17	20	18
8	–	3	149	52	183	9
9	–	3	78	34	14	34
10	–	6	66	46	70	8
11	–	5	89	55	48	11
12	–	6	84	56	31	12
Sulitjelma Gabbro complex						
2	–	3	105	47	22	27
3	–	3	103	26	10	42
5	N	3	21	−12	26	25
10	N	7	24	−13	58	8
12	N	3	21	−12	26	25
Lyngen Ophiolite complex						
1	R	3	105	−12	94	13
3	–	4	36	57	36	16
5	–	6	73	31	21	15
8	R	3	162	0	18	30
10	–	4	289	47	12	27
15	N	3	306	10	29	23

* Mean directions are only listed here for directions defined by three or more samples from the same site. Group mean directions at a sample level are given in Table 2. D and I are the mean declination and inclination derived from N samples with precision estimate, k $[=(N-1)/(N-R)]$ and cone of confidence of α_{95} about the mean direction.

from this pluton yield directions of magnetisation removed from the post-310 Ma palaeofield migration path for the area (Fig. 5), and it is therefore probable that acquisition of the "A", "B", "C" components dates from between the peak of tectonic activity (ca. 430–420 Ma) and the Siluro-Devonian uplift history (ca. 410–390 Ma).

The Sulitjelma Ophiolite

The Sulitjelma complex (16°E, 67.2°N) displays the characteristic structural sequence of gabbros, sheeted dyke complex and pillow lavas constituting the upper part of an ophiolite (Mason, 1980; Boyle, 1980). Geothermobarometric estimates from coexisting cordierite and garnet in contact hornfelses yield estimates of 6.5 kbar (Mason, 1980), consistent with values for the pressure at the peak of metamorphism in the sedimentary rocks deposited above the ophiolite. The gabbros have been sheared and metamorphosed in a wide zone along their southern margin, giving rise to flaser gabbro (Mason, 1971) and are tectonically inverted (Boyle et al., 1979). Interpretation of the pillow lava and sheeted complex members of the ophiolite as continuous with, and grading into, the gabbros casts doubt upon major thrusts proposed by some early studies (e.g., Kautsky, 1953) and still defended by some present-day workers (Stephens et al., 1985a). The Furulund Granite, an intrusive into the same country rocks as the gabbro, has been dated (Rb–Sr) at 421 ± 11 Ma (Wilson, 1981). Other dated syntectonic igneous intrusions in the adjoining area are the Baldoaivve

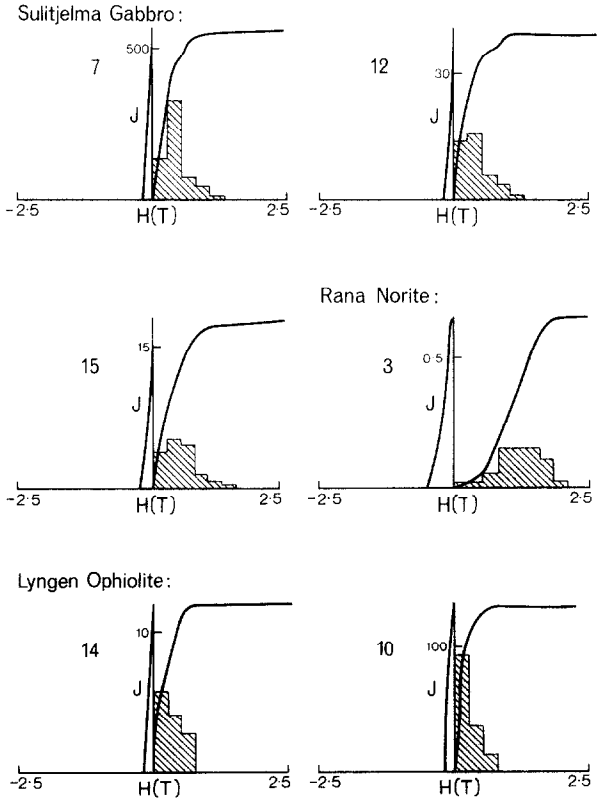


Fig. 4. Some representative IRM determinations on palaeomagnetic samples of this collection.

dykes (422 ± 8 Ma) and the Duoldagop dyke (418 ± 14 Ma, Wilson, 1981). Fossils of Late Ordovician to Early Silurian age are found at the top of the flysch carapace covering the ophiolite and yield a minimum age for its origin, and it seems likely that the Sulitjelma Gabbros are of similar age to the geochemically and petrographically-similar Råna Norite intrusion (next section), dated 437 ± 2 Ma (Barnes 1990).

Although the gabbros of the Sulitjelma ophiolite show variable degrees of iron enrichment, most of the iron is present in pyroxenes and olivines, and microscopically-visible opaque phases are sparse; they comprise mostly pyrrhotite and ilmenite (Mason, 1971). The regional metamorphism has produced coronas between olivine and plagioclase, and subsequent hydration has produced green hornblende and clinozoisite; it seems most likely that the remanent magnetism is linked to ferromagnetic minerals formed by recrystallisation during this metamorphic event.

The northwest lobe of the gabbro has been sampled at fifteen sites located along a transect (Fig. 2b) which extends a study by Piper (1974) on the southwestern amphibolitised margin of the body. Seventy-six percent of these samples defined components of magnetisation meeting acceptability criteria and directions fall into three groups (Fig. 5). A well defined group of NNE-directed magnetisations is complemented by a more disper-

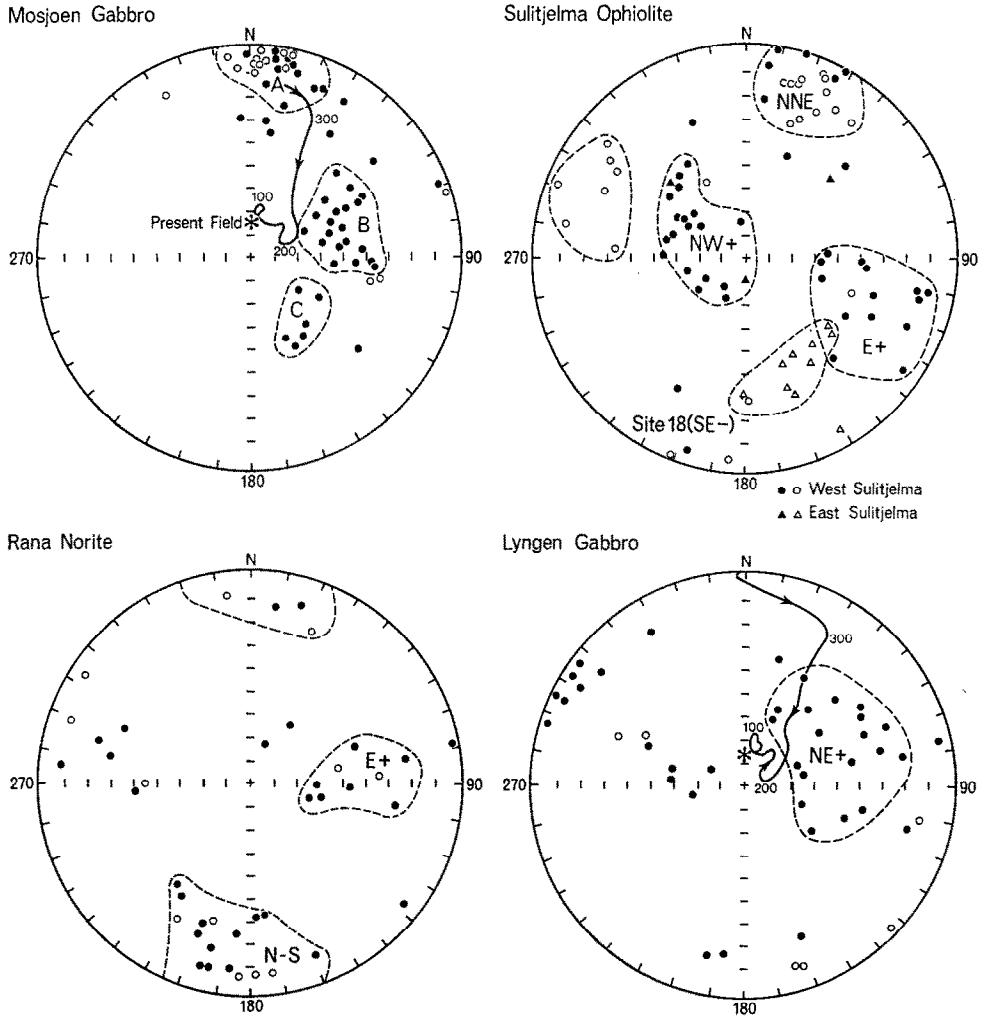


Fig. 5. Directions of magnetisation of components identified in the Møsjoen, Sulitjelma, Råna and Lyngen bodies. Solid symbols are lower hemisphere plots and open symbols are upper hemisphere plots; equal area projections.

sed group of five antiparallel directions. This component is represented by samples with two types of magnetic characteristics: firstly it resides in samples with a strong ($5.0–1.8 \times 10^{-5}$ A m^2 kg^{-1}) remanence in most samples from site 10 and in isolated cores from other sites subtractable by a.f. cleaning but with moderate to high coercivities ($J_{20}/J_0 = 0.8–0.2$, Fig. 6); it is not quite discrete and a slightly steeper north-northeast component with lower blocking temperature/coercivity is first

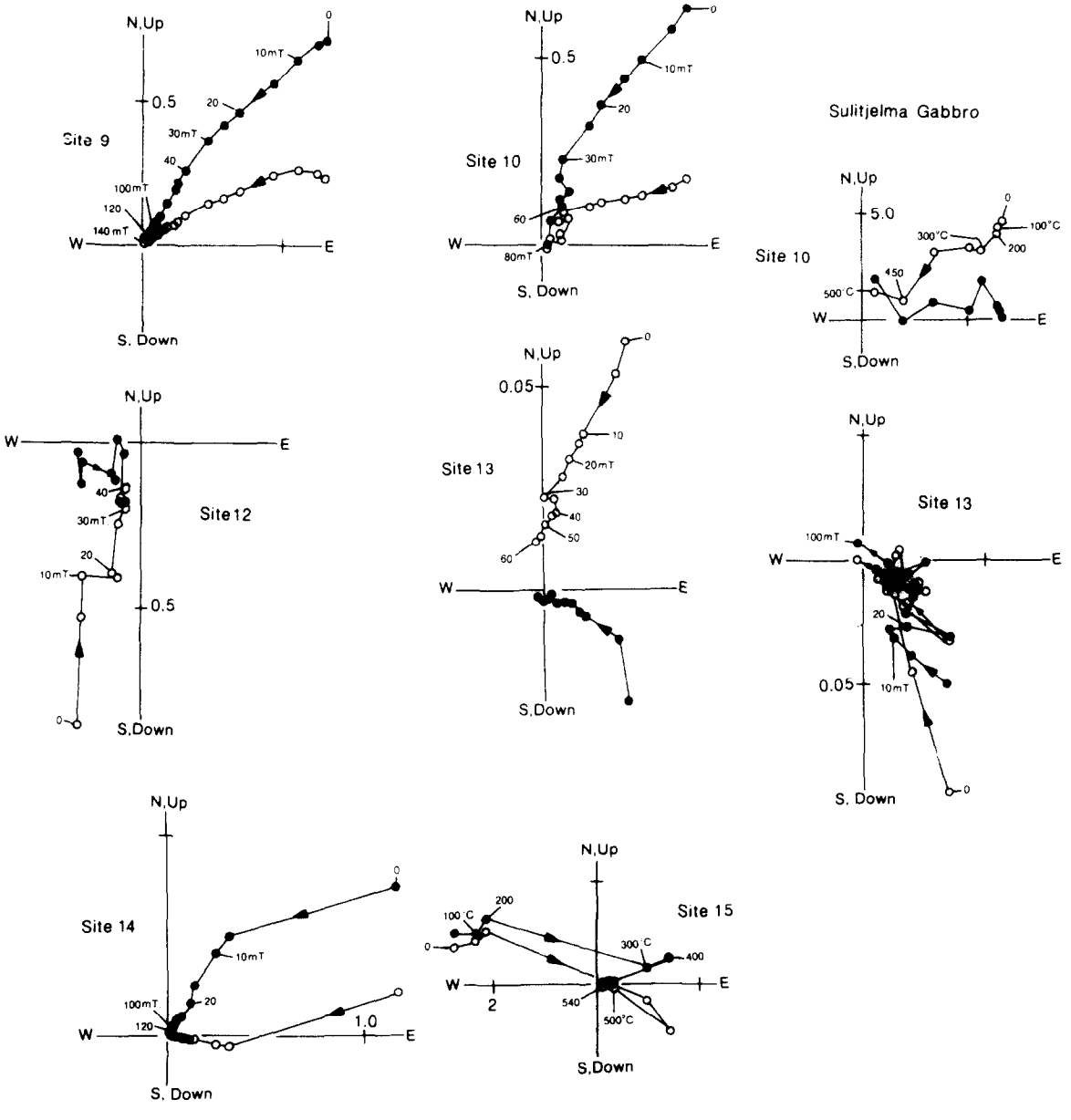


Fig. 6. Representative components of magnetisation in the Sulitjelma Gabbro complex. The most common components recognised in this complex have N to NNE declinations with shallow inclinations (sites 9, 10, 13 and 14); they are often not discrete and move from NNE to N directions at higher stages of cleaning (sites 9 and 10). Site 13 also illustrates an example of the E positive group and site 12 is an example of the westerly steep positive group. Other components sporadically recorded, mostly in isolated samples, have easterly shallow to negative directions (site 10); in the sample from site 15 this appears to be recovered following subtraction of a low blocking temperature westerly negative component. Symbols are as for Fig. 3.

subtracted to yield a more northerly component with shallower inclination (Fig. 6). Samples with this property tend to have stepped IRM curves (Fig. 4) suggesting that more than one phase is carrying the remanence. This direction is otherwise represented as a weaker ($0.15–0.01 \times 10^{-5}$ A m^2 kg^{-1}) component in a minority of samples from most other sites.

A more dispersed group of east-southeast shallow components is patchily represented at most sites along this traverse as a fairly weak components (0.18–0.01 A m^2 kg^{-1}); in some cases it is little influenced by a.f. treatment (site 13) but is recognised by thermal cleaning and subtracted below the Curie point of magnetite (sites 10 and 15 in Fig. 6). Whilst there is evidence for components with easterly and westerly declinations residing in more than one phase, they have survived only sporadically. A third group of samples yields steeper northwest or westerly positive directions (e.g. 12 in Fig. 6); they are patchily represented at all sites and are usually resident in stronger components (1.8–0.2 A m^2 kg^{-1}) subtractable by a.f. cleaning. These samples yield steep IRM curves and have lower coercivities than the north shallow group.

Group mean directions of magnetisation calculated from the sample groups illustrated in Fig. 5 are listed in Table 2. The north shallow direction identified in this study is close to a component derived from an earlier study in the southwest sector of the gabbro (Piper, 1974; $D = 15°$, $I = 15°$), but is typically a more stable remanence in these interior sites than in the amphibolitised margin. Although separation of these components is not sufficient to merit more than one mean calculation, their dispersal is probably a reflection of their composite origin (as is observed in individual samples from site 10). A similar behaviour is identified in some samples of the Møsjoen Gabbro (see examples from 1 and 2 in Fig. 3); it may reflect a protracted acquisition as the ambient field direction changed during retrogressive metamorphism and uplift-related cooling.

The eastern part of the ophiolite complex has been sampled at six sites in the Otervatn Volcanic

TABLE 2

Palaeomagnetic results, gabbro intrusions and ophiolite complexes, Norwegian Caledonides

	D (°)	I (°)	N	R	k	α_{95} (°)	Palaeomagnetic pole		
							°E	°N	*dp/dm*
Møsjoen Gabbro (13.3°E, 64.9°N)									
A(N)	7	3	23	22.4	38	5.0	186	25	2.5/5.0
B(E +)	74	49	19	18.5	41	5.3	105	34	4.7/7.0
C(SE +)	140	51	7	6.8	33	10.7	48	12	9.8/14.5
Sulitjelma Gabbro (16.2°E, 67.2°N)									
West (N–S axis)	20	−9	22	21.1	23	6.6	175	17	3.4/6.7
West (+E)	110	29	20	18.8	15	8.6	82	7	5.2/9.5
West (NW +)	302	64	17	15.9	14	9.8	280	53	12.3/15.5
East (SE −)	155	−35	10	9.5	19	11.4	47E	−40	7.5/13.1
Råna Norite complex (17.0°E, 68.3°N)									
(N–S Axis)	9	−16	19	18.0	19	7.9	188	13	4.2/8.2
(E +)	101	37	11	10.0	11	14.9	89	15	10.3/17.5
Lyngen Ophiolite (20.3°E, 69.8°N)									
(NE +)	65	43	19	17.4	11	10.6	124E	33N	8.2/13.2

All results are quoted in situ. D and I are the mean declination and inclination derived from N samples where R is the resultant vector, k is the precision parameter [$= (N-1)/(N-R)$] and α_{95} is the radius of the confidence circle about the mean direction. *dp* and *dm* are the radii of the 95% confidence oval about the pole position in the colatitude direction and at right angles to it respectively.

Formation (sites 16, 21 and 20), gabbros within the Hammaren shear zone (sites 17 and 19, Fig. 2b), and in the massive gabbros (site 18). The volcanic facies of the ophiolite yielded no useful information, but stable components were resolved from sites 17 and 18. The gabbro site (18) defined a coherent south-southeast negative direction progressively subtracted by a.f. cleaning (Fig. 7) and contrasting with components resolved from the western part of the gabbro outcrop (Figs. 5 and 6). Results from the sheared gabbros of site 17 are more variable; components with directions comparable to the western part of the complex are represented here and all appear to be resident in magnetite (Fig. 7).

The Råna Norite

The Råna Norite intrusion is emplaced into gneisses of the Narvik Group east of Ofolen Fjord at 17.0°E, 68.5°N (Fig. 2). It predates the third phase of deformation in this area and appears to be contemporaneous with amphibolite facies metamorphism of the country rocks (Boyd and Mathiesen, 1979). The igneous facies have a concentric distribution and a conformity with the country rock foliation (Fig. 2): there is a gradational contact between marginal norite and quartz norite with bodies of various ultrabasic rock types concentrated towards the margin. Foslie (1921) regarded the intrusion as the product of fractional crystalisation and magmatic squeezing during the syn- to late-tectonic history. Gustavson (1969) stressed the importance of gravitational settling with the heavier magmatic segregations now concentrated around the margin of the outcrop; although now deformed along an E–W fold axis, the shape of the body approximates to an inverted cone (Stephens et al., 1985b).

The palaeomagnetic sample, comprising a N–S traverse of nine sites, comes predominantly from the interior quartz-norite facies (Fig. 2c). Opaque phases are rare and found only as minute inclusions in olivine and secondary talc (Foslie, 1921). Uralitisation of the pyroxenes with talc and antigorite rims and fractures in the olivine and pyroxene grains record metamorphic retrogression; the ferromagnetic content is probably linked to this

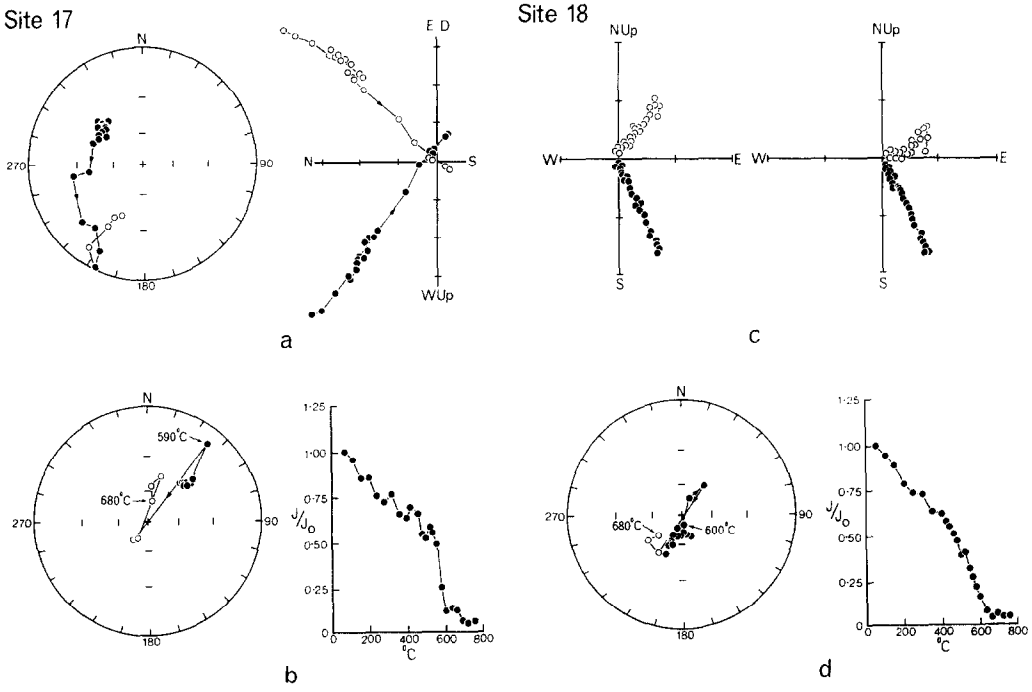


Fig. 7. Components of magnetisation recovered from a.f. and thermal demagnetisation of samples from the eastern margin of the Sulitjelma Gabbro Complex. Symbols are as for Fig. 3.

metamorphism. The remanence has fairly high coercivities with IRM curves continuing to rise slightly in higher saturating fields indicative of hematite (Fig. 4). The natural remanences are weak ($1.15–0.01 \times 10^{-5}$ A m^2 kg^{-1}). Some components can be progressively subtracted by a.f.

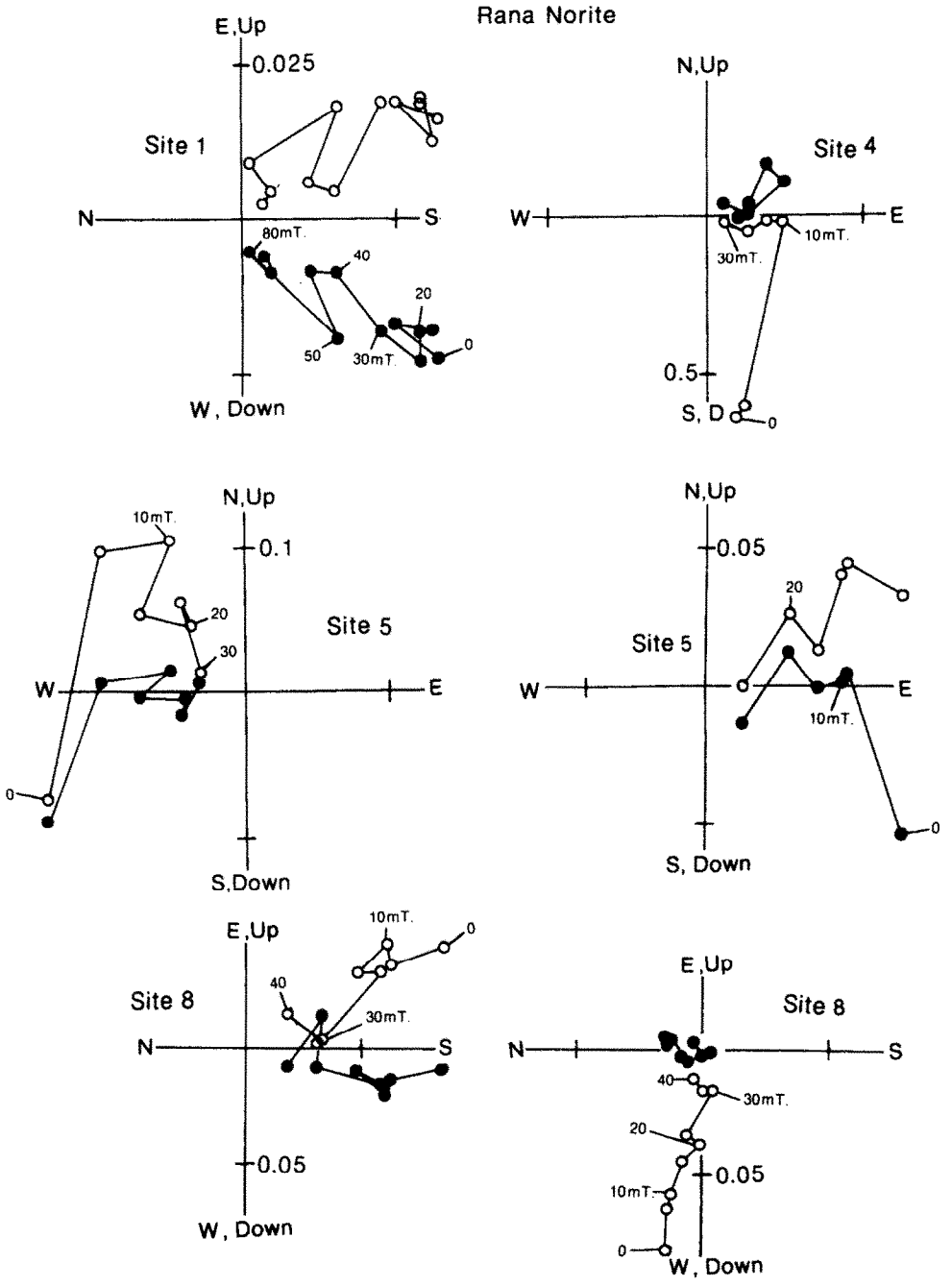


Fig. 8. Components of magnetisation resolved by a.f. demagnetisation in samples from the Råna Norite. No components were successfully demagnetised by thermal cleaning in this collection and no single direction is prominently represented at any one site. Sites 1 and 8 illustrate southerly shallow negative components (see also Fig. 5). Site 4 is an example of an easterly shallow component; other isolated samples have similar declinations with positive inclinations or near-anti-parallel equivalents (site 5). One sample from site 8 shows an example of a probable present field component. Symbols are as for Fig. 3; sampling sites refer to Fig. 2.

cleaning; others are little affected by this technique although thermal demagnetisation was unsuccessful in defining components at higher fields with erratic increases in intensity often observed above 300 °C. Prominent components recognised by a.f. cleaning have N- and S-directed remanences with shallow inclinations (e.g., sites 1 and 8 in Fig. 8). Collectively they define a N–S group with a mean of $D = 9°$, $I = -16°$ ($N = 19$ samples, $\alpha_{95} = 8°$). Contrasting directions are sporadically represented at sites 1 to 7 (Fig. 5). They include easterly shallow-directed magnetisations with three possible antiparallel members grouped as component "B" in Table 2 and present at sites 4 and 5–8; they are subtracted by low a.f. fields (sites 4, 5 and 8, fig. 8). In view of their poor definition they are not considered further here.

The Lyngen Gabbro

The Lyngen nappe (20 °E, 69.5 °N) comprises three major divisions: the lowest is the Koppangen Formation, which is succeeded by the Lyngen Gabbro and Kjøsen Formation (with the latter represented by deformed amphibolites). The uppermost and youngest division, the Balsfjord Group, is palaeontologically dated as Late Ordovician–Early Silurian (Andresen et al., 1985). The gabbro complex forms the upstanding axis of the Lyngen Peninsula (Fig. 2d) and has been interpreted as the cumulate part of a dismembered ophiolite (Minsaas and Sturt, 1985). Whilst the Koppangen and Kjøsen formations have been metamorphosed up to high amphibolite grades, the gabbro and higher units are characterised by pervasive greenschist facies metamorphism and are extensively uralitised and saussuritised (Randall, 1971a). Three episodes of folding include an early recumbent episode, major open folds, and late cleavage formation linked to minor upright folds. An early extensional phase is recorded by metadolerite dykes; related to the second episode of folding and influenced by the third. More recent studies (Andressen et al., 1985) have not supported the presence of thrust faults on the large scale indicated by earlier workers (e.g. Randall, 1971b).

The palaeomagnetic sample comprises a section through the central part of the complex at Kjøsen Fjord, four metadolerite intrusions in the overlying Balsfjord Group (sites 7–10) and a subsidiary section on the south eastern part of Sjør Fjorden (site 14–17, Fig. 2d). The pervasive greenschist facies metamorphism has had a profound affect on the magnetic properties of the gabbro and less than 50% of the collection yielded useful palaeomagnetic information. Instability of magnetisation correlates with steep IRM curves (Fig. 4) indicative of the presence of multidomain magnetite or pyrrhotite carriers, and progressive increase in susceptibility with temperature also indicative of multidomain titanomagnetites (Piper, 1987). Samples yielding stable components of magnetisation have higher coercivities as indicated by IRM studies (Fig. 4) and either weak susceptibilities showing little change with temperature, or higher susceptibilities showing progressive reduction with increasing temperature and linked to single domain carriers in magnetites subdivided by ilmenite lamellae. Positive N- to NE-directed components of magnetisation are prominently represented in these rocks and typically subtracted below 300 °C and in low a.f. fields (Fig. 9). They tend to have declinations plotting somewhat to the east of the Mesozoic–Recent field and are represented at sites 3 to 5 and 9 to 11. One site (5) yielded a single coherent component of magnetisation ($D = 73°$, $I = 31°$, $\alpha_{95} = 17°$, $N = 7$). No single magnetisation is represented at other sites although a fairly dispersed group of NE positive-directed components (Figs. 5 and 9) resides in strong to weak magnetisations ($25–0.03 \times 10^{-5}$ A m^2 kg^{-1}) subtracted by a.f. cleaning with moderate to low coercivities ($J_{20}/J_0 = 0.6–0.05$) and below the Curie point of magnetite (Fig. 9).

Interpretation

The principal feature to emerge from study of three of the four basic igneous complexes in this study is a shallow N–S dipolar axis of magnetisation. Since this is common to bodies in differing local tectonic settings, it must postdate major Caledonian tectonism. In all these areas synorogenic events linked to the Scandian tectonic epi-

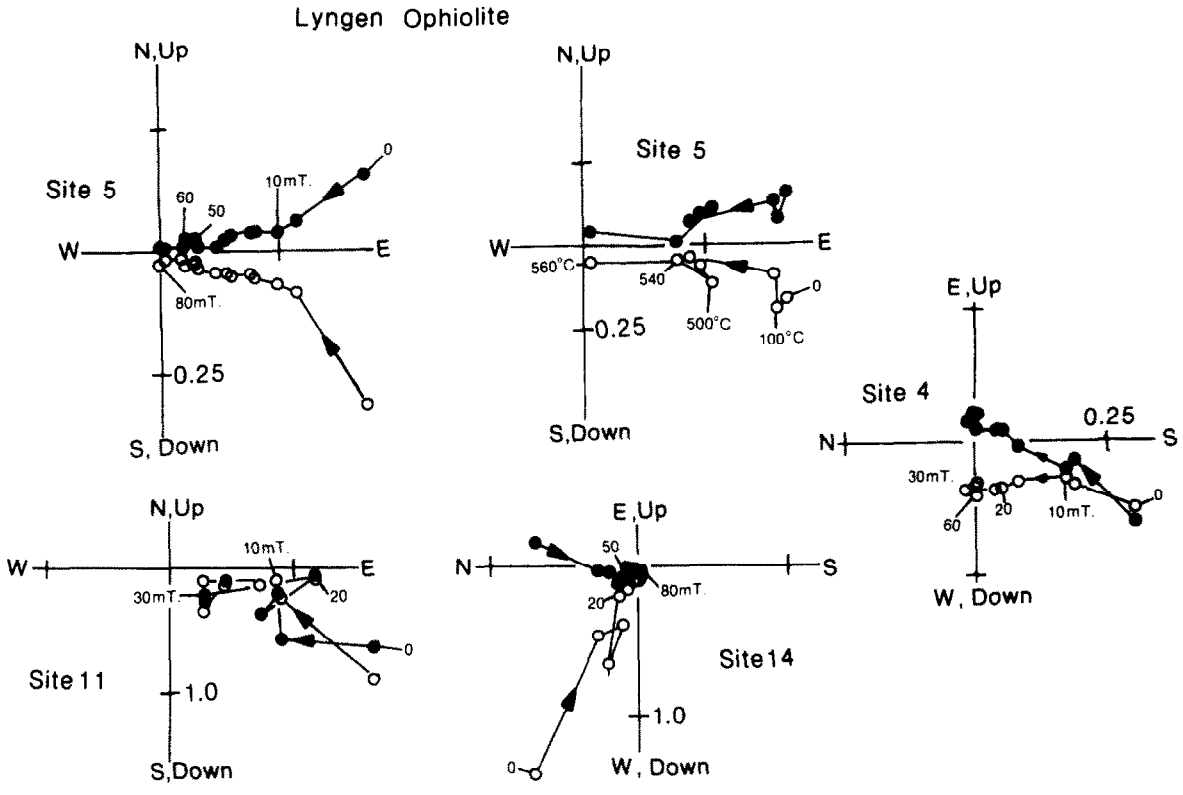


Fig. 9. Components of magnetisation resolved by a.f. and thermal demagnetisation at sites in the Lyngen Ophiolite body. Site 5 examples illustrate identification of the easterly positive component by a.f. and thermal demagnetisation; isolated samples showing comparable components are found elsewhere (site 11). Site 4 is an example of a shallow southerly component subtracted by a.f. cleaning leaving a residual component not resolved by this method and site 14 is an example of a probable Recent field component. Symbols are as for Fig. 3 and the site numbers refer to Fig. 2; sites from which significant palaeomagnetic information has been derived are shown as closed symbols in this latter figure.

sode are dated ca. 420 ma and the magnetisations must therefore be younger than this. Since Lower Devonian molasse facies sediments occupy the infrastructure of the orogen in southwestern Norway, it is probable that these bodies had attained high levels following orogenic collapse by Early Devonian times. Hence if this remanence is to be linked to post-orogenic uplift and cooling, it was most probably acquired during the interval 420–400 Ma.

For comparitive purposes the palaeomagnetic results from the British Caledonides (Briden et al., 1984; Piper, 1987) currently provide the best-constrained pattern of Silurian–Carboniferous APW for the European Block (Fig. 10). Since the Eurasian and Laurentian forelands were essentially sutured by these times (see, for example, Phillips et al., 1976, and Roberts and Gale, 1978), it is logical to expect the essential features of this path to emerge from continental Europe. In the event present results from Ordovician–Carboniferous rocks of continental Europe only identify a small segment, essentially comprising the Late Carboniferous–Permian part, of this path (Piper, 1987). In addition, they are predominantly reversely magnetised suggesting that widespread remagnetisation has taken place here during the Permo-Carboniferous reversed (PCR) superchron.

A distinctive feature of the British path is the westerly location of results from a wide range of Old Red Sandstone lavas and sediments (Chamalaun and Creer, 1964; Sallomy and Piper, 1973a; McClelland Brown, 1983; Piper, 1987). Field tests (Sallomy and Piper, 1973a; Torsvik, 1985a) imply that these magnetisations are pre-mid-Devonian in age. This is well defined in results from northern Britain bordering, and within, the orthotectonic Caledonides and is the continua-

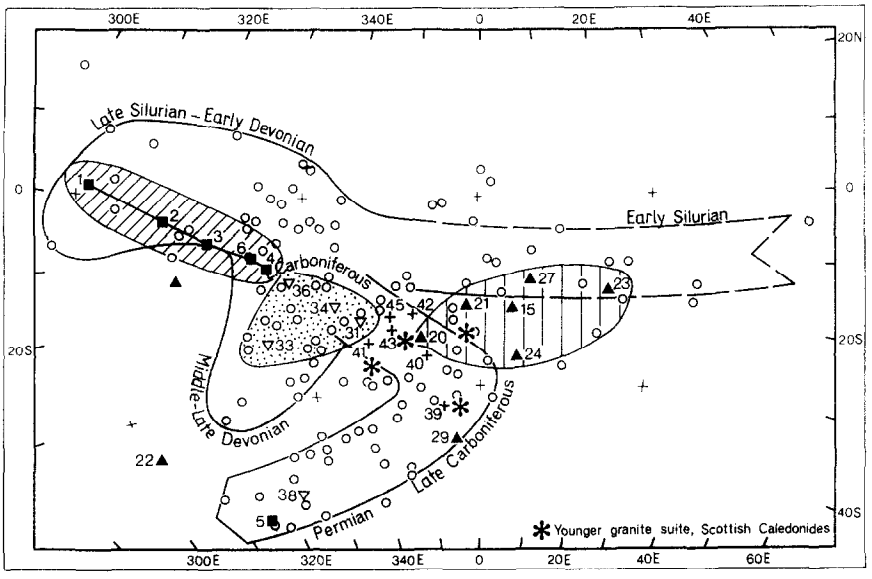


Fig. 10. Palaeomagnetic poles from the Scandinavian Caledonides and the adjoining cratonic foreland of the Baltic platform compared with the Silurian–Permian APW path derived from the British Isles. The latter path, shown in outline form here, is defined by the poles illustrated as open circles and is more fully discussed elsewhere (Briden et al., 1984; Piper 1987; Torsvik et al., 1988). The poles from the Scandinavian Caledonides and the adjoining Baltic Platform are numbered as follows: *1–4* = Jotun nappe metamorphic rocks, Caledonian uplift remanence, groups 1 to 4 (Piper and Poppleton, 1990); *5* = Fongen–Hyllingen gabbro SW component, Silurian ca. 405 Ma (Rb–Sr, Abrahamsen et al., 1979); *6* = Basal Gneiss region NE + /SW − post-Caledonian uplift components (Piper and Poppleton, 1990); *7* = Sulitjelma gabbro N–S shallow axis (this study); *8* = dolerites, Sarv nappe, 702 Ma (Rb–Sr, age of intrusion?) Caledonian overprint (Bylund and Zellman, 1980); *9* = Roysing dyke, Sarv nappe (Bylund and Zellman 1980); *10* = Møsjoen Gabbro N–S shallow axis (this study); *11* = Rana Norite N–S shallow axis (this study); *12* = metagabbros, Lokken, Caledonian uplift remanence (authors' unpublished result); *13* = gabbro, Ullsberg, Caledonian uplift remanence (authors' unpublished result); *14* = Roragen Sandstones, Early Devonian age of sedimentation, Late Devonian–Early Carboniferous (Solundian) overprint (Storetvedt and Gjellstad 1966); *15, 16* = Kvamshesten sediments with result 16 including Dalsfjord mylonites, Middle–Late Devonian sedimentary age, Late Devonian–Early Carboniferous overprint (Lie et al., 1969; Torsvik et al., 1986); *17* = Hasteinen sediments, Late Devonian–Early Devonian overprint (Torsvik et al., 1987); *18* = Hornelen sediments, Early Devonian, Early Carboniferous overprint (Torsvik et al., 1988); *19* = Vanern limestones, Ordovician rock age of sedimentation,? Late Palaeozoic overprint (Mulder, 1965); *20–23* = Gotland limestones, Silurian age of sedimentation,? Late Palaeozoic overprint (Claesson, 1979); *24* = Ringerike Sandstone, Late Silurian–Early Devonian age of sedimentation, pre-Mid-Carboniferous remanence (Douglass, 1988); *25* = Lyngen Gabbro, in situ remanence (this study); *26* = Basal Gneiss region, Late Palaeozoic–Mesozoic overprint (Piper and Poppleton, 1989). Poles shown by solid squares are derived from the Caledonian plutonic terrane of South Norway; solid upright triangles are poles from basic bodies in the central Caledonides; inverted open triangles are from late to post-tectonic molassic sediments and define a group shown by the stippled area; and crosses are poles from the Baltic platform; the latter are mostly from rocks of Early Palaeozoic age but yield pole positions falling on the Carboniferous paths and are therefore likely to be overprints of this age.

tion of a long latitudinal swathe recognised in uplift and cooling magnetisations from plutonic rocks in this sector of the orogen (Esang and Piper 1984). The central part of the swathe is recorded by members of the "younger granite" suite in the Scottish Caledonides (Torsvik 1985b) linked to age dates in the range 430–414 Ma. Table 3 is a summary of the palaeomagnetic results linked to slow uplift-related cooling in the orthotectonic Caledonides of Scotland. They include results from both the Northern Highland and the Grampian Highland divisions on either side of the Great Glen Fault, and span the full exposed width of the orogen here from Peterhead in the east to Ross of Mull in the west. In view of the definition of this N–S shallow axis by the Silurian "younger granite" suite, following Torsvik (op. cit.) we are inclined to regard similar magnetisations from the metamorphic Caledonides as the products of late Caledonian thermochemical overprinting. This

distribution includes results formerly interpreted as Ordovician such as from the "younger" gabbros of Aberdeenshire (Sallomy and Piper, 1973b; Watts and Briden, 1984). The source of the overprinting may be linked to widespread thermal effects of the "younger granite" and early Old Red Sandstone magmatism. The continuation of the path to the westerly position defined by most of the "Old Red Sandstone" results is recorded by a change in palaeofield directions from N–S shallow to NE–SW intermediate noted in the introduction. It is recorded (poles 1–4 and 6 in Fig. 10) by uplift and cooling magnetisations from the Basal Gneiss Region and Jotun nappe in the southern sector of the Scandinavian Caledonides (Piper and Poppleton, 1990) and implies that the Caledonides collectively moved from low paleolatitudes (0–12°S) to intermediate latitudes (45–55°S) between mid-Silurian and Early Devonian times.

The return path to the Carboniferous location in Fig. 10 is still poorly defined and currently there are discrepancies between pole positions derived from Lower Carboniferous rocks. Results from volcanics of this age in Derbyshire and Scotland (Piper, Atkinson and Norris, in prep.) confirm a pole position just 10–15° south of the Upper Silurian–Lower Devonian group and first identified by Wilson and Everitt (1963). Carbonates of the same age yield poles plotting 25° further to the southeast and close to the Upper Carboniferous–Permian segment in Fig. 10; the limestones show reversals of magnetisations (see for example Turner and Tarling, 1975) and cannot therefore have been comprehensively remagnetised during the PCR superchron although there must be a suspicion that the directions are contaminated by a Variscan overprint. Clarification of the Early Carboniferous pole position identifies a large APW shift during mid-Carboniferous times in a southeasterly sense (on the (south) APW path illustrated in Fig. 10) and a common movement is implied by refined analysis of North American results of this age (Roy and Morris, 1983).

From correlation with the British and European APW path (Fig. 10) the N–SW shallow-directed components identified in the basic bodies of this study could be interpreted either (1) as an uplift

TABLE 3

Palaeomagnetic results from Caledonian plutonic terranes interpreted to record late Caledonian (Middle–Late Silurian) thermochemical events

Rock unit	Remanence age (Ma)	N(n)	k	α_{95}	Pole position (reversed)					Reference
					D (°)	I (°)	°E	°S	dp/dm (°)	
(1) Scottish Caledonides, Northern Highlands										
Helmsdale Granite	420 [b], 400 [c]	(17)	32	6.0	181	2	355	31	3.9/3.0	Torsvik et al. (1983)
Strontian Granite	421–381 [c]	(41)	18	5.2	190	23	344	21	2.9/5.5	Torsvik (1985b)
Ross of Mull Granite	414 [a]	(36)	9	8.3	193	13	339	26	4.3/3.5	J.D.A. Piper, unpubl. result
Microdiorite–Appinite Suite	420–400 [d]	41	14	6.2	187	29	346	17	3.8/6.9	Esang and Piper (1984)
Dolerite Suite	420–400 [d]	22	10	10.1	188	32	348	15	6.4/11.4	Esang and Piper (1984)
(2) Scottish Caledonides, Grampian Highlands										
Peterhead Granite	415 [a]	(38)	19	5.2	180	22	358	21	2.9/5.5	Torsvik (1985b)
Foyers Granite	404–392 [c]	(41)	15	5.7	188	9	347	27	2.9/5.8	Torsvik (1985b)
(3) Central Norwegian Caledonides										
Møsjoen Gabbro	< 420, > 390(?)	(23)	38	5.0	187	−3	6	25	2.5/5.0	this study
Sulitjelma Gabbro	< 418, > 390(?)	(22)	23	6.6	200	9	355	17	3.4/6.7	this study
Råna Norite	< 418, > 390(?)	(19)	19	7.9	189	16	8	13	4.2/8.2	this study

[a] Rb-Sr, [b] U-Pb, [c] K-Ar, [d] Regional K–Ar "chrontour" ages. D and I are mean declination and inclination derived from N(n) palaeomagnetic sites (samples) with a precision parameter k and cone of 95% confidence equal to α_{95}, dp and dm are the radii of the 95% confidence oval about the mean direction in the colatitude direction and at right angles to it respectively.

and cooling remanence synchronous with, or marginally older than, the remanence in the "Younger granites" of the Scottish Caledonides, or (2) as a mid-Carboniferous overprint. The latter possibility can be excluded on two counts. Firstly, the poles plot mostly to the east of the mid-Carboniferous path and show only marginal overlap with data of this age (Fig. 10). Secondly, the poles from this study plot to the east of a suite of poles from Devonian sedimentary basins bordering the western margin of the orogen in south Norway (Torsvik et al., 1986, 1987, 1988). These poles (from the Kvamshesten, Hasteinen and Hornelen formations) plot as a discrete population close to the Early Carboniferous position for Britain (Fig. 10). This observation and their geological history argue for resetting of the remanence accompanying burial and lower greenschist facies metamorphism in the Solundian (Svalbardian) tectonic pulse here at ca. 375 Ma (Torsvik et al., 1988). A component with a comparable direction in the Ringerike Sandstone on the Baltic platform predates mid-Carboniferous folding (Douglass, 1988). The contrasting remanence (Fig. 10) recorded in the basic bodies from the central Caledonides (vertically-shaded area in Fig. 10) is likely to be appreciably older than this group. If acquired during uplift-related cooling it would predate truncation of the Finnemarkian and Scandian structures and development of the marked angular unconformity. This requires deep Siluro-Devonian erosion prior to deposition of these Devonian sedimentary sequences and suggests that the characteristic remanence in the basement was acquired prior to 390 Ma. If this is the case, the east to west polar movement recorded by the basement and cover rocks is the analogue of the identical E–W shift recognised in the British Caledonides (Fig. 10). In fact the pattern of APW is probably somewhat more complex than this in detail: the Late Devonian to Early Carboniferous poles from Scandinavia appear to be part of a SE trending *return* path continuing the Early Devonian extension of the path recognised in Britain and since found in Scandinavia: the SW + component in the Fongen–Hyllingen body yields a pole plotting on this westerly extension (Abrahamsen et al., 1979), and a comparable NE – /SW + axis is also defined by the predominant remanence in the Basal Gneiss Region and Jotun nappe complex from the southern sector of the Scandinavian Caledonides (poles 1–4, and 6, inclined shaded area in Fig. 10, Piper and Poppleton, 1990).

The Fongen–Hyllingen pluton has the southernmost location of the palaeomagnetically-studied basic bodies in the Scandinavian Caledonides. Hence there is evidence here for a transition in the uplift-related remanence record along the length of the orogen: in the southern sector the westerly Early-Devonian loops the major feature recorded by the plutonic rocks (Piper and Poppleton, 1990), whilst in the central sector the easterly precursor path (this paper) is the main feature to be currently identified. This would be compatible with the earlier completion of tectonism in the NNE trending central part of the belt in Scandinavia (Roberts and Gale, 1978; Roberts and Gee, 1985).

We therefore interpret the predominant axis recognised in the basic bodies of this study as a magnetic imprint acquired during uplift-related cooling in Middle to Late Silurian times and contemporaneous with the widespread late Caledonian thermochemical event recorded in the Scottish Caledonides (cf. Tables 2 and 3). The other components recognised in this study are more difficult to interpret. Two poles (Lyngen NE + and Mosjoen E +) plot fairly close to the palaeofield migration path at ca. 200 Ma (Fig. 11) and there must be a suspicion that they record an overprint of Mesozoic age. This possibility is suggested by identification of magnetic overprints of this general age at a number of locations around the North Sea basin (e.g., Turner, 1977) and along the coastal margin of Norway (Turner et al., 1978; Storetvedt et al., 1978, Løvlie and Mitchell, 1982; Torsvik et al., 1988). They are shown by the crosses in Fig. 11 and probably result from thermal consequences of tectonic developments in the North Sea and Atlantic-bordering grabens. This explanation is not entirely satisfactory because the Lyngen and Mosjoen components have declinations to the east of the predicted path (Fig. 9). It is possible (1) that the APW path has more ramifications than the mean track illustrated in Fig. 11 and/or (2) that the crustal section has been sub-

ject to flexure since acquisition of a secondary remanence. A small seaward flexure linked to basin development in the North Sea is a possible explanation for divergence of the characteristic remanence in the Karmøy ophiolite from the Mesozoic path at 210–170 Ma (Poppleton and Piper, 1990, and Fig. 11). The Lyngen NE + component would require an appreciable westward tilt of 25° to be statistically comparable with the Mesozoic path; whilst a tilt in this sense could help to explain the present outcrop at Lyngen where the nappe pile dips to the west (Andressen et al., 1985), the location remote from the continental margin cannot be reconciled with the consequences of post-Palaeozoic extension.

Hence we are forced to consider that these and other components recognised by this study may be relict magnetisations which have escaped full overprinting by the Caledonian (mid-Silurian) thermochemical event. Continent–continent collision along the site of the orogen was essentially completed in late Ordovician times with the imbrication and translation of the nappe pile continuing through Silurian into Early Devonian times (see for example, Roberts and Gee, 1985; Gee et al., 1985; Stephens et al., 1985a). Although the predominant remanences recognised in this and other studies are essentially post-tectonic phenomena, there was a protracted Caledonian history during which any relict components could have been subject to deformation. This has been argued for the Sarv Nappe "A" and Fongen–Hyllingen "A" components by Abrahamsen et al. (1979) and Bylund and Zellman (1980), respectively. The possibility of deformation precludes interpretation of individual results, but we observe here that collectively they all plot along a precursor path to the Siluro-Devonian path shown in Fig. 10 and described above (see Fig. 11). Results from elsewhere in the orthotectonic Caledonides plot along this path including poles from the Karmøy ophiolite in southwest Norway (Poppleton and Piper, 1990); the Basal Gneiss terrane in southwest Norway (Piper and Poppleton, 1990), and the Unst Ophiolite in Shetland emplaced onto the opposite foreland of the orogen (Taylor, 1988). Since the latitudinal distribution of these poles is compatible with the consequences of tectonic deformation (Mac-

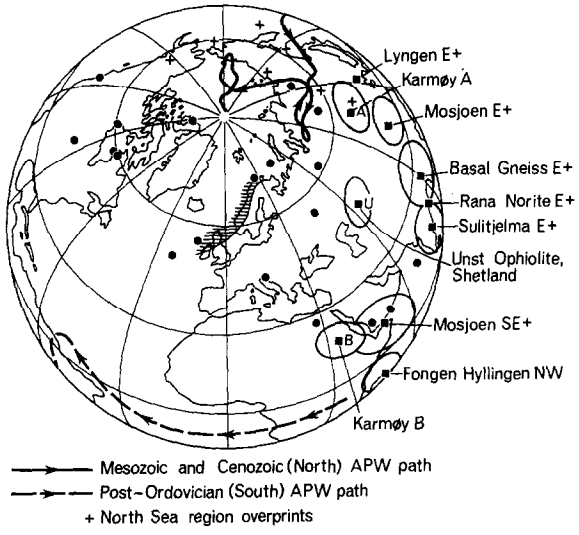


Fig. 11. Possible relict magnetisations from the orthotectonic Caledonides of Scandinavia and Scotland (Shetland) recognised by this and other studies. Note that these poles all plot to the east of the Mesozoic APW path and overprints acquired in the vicinity of the North Sea Basin (crosses). They suggest a precursor to the Siluro-Devonian path indicated here as a south APW path and shown in Fig. 10. The closed dots are probable Late Ordovician poles (Perroud and Van der Voo, 1985; Poppleton and Piper, 1990) indicative of a high (south) latitudinal location for Europe during these times. The poles from the orthotectonic Caledonides plotted with 95% confidence ovals are from this study, pole U from Taylor (1988), Fongen Hyllingen NW (Abrahamsen et al., 1979) Karmøy results (Poppleton and Piper 1990) and Basal gneiss terrane of southern Norway (Piper and Poppleton, 1990).

donald, 1980), it would be premature to use them to define an APW path. However, identification of poles from the same tectonic divisions in southwest Norway (Karmøy A and B) and central Norway (Møsjoen E+ and SE+) plotting at the extremities of this distribution, strongly suggests that it includes a large measure of APW. The recognition and consequences of large APW movement in Late Ordovician and Early Silurian times have been discussed elsewhere (Perroud and Van der Voo, 1985; Piper, 1987; Poppleton and Piper, 1990).

Acknowledgements

T.J. Poppleton and A.J. Griffiths acknowledge receipt of NERC studentships and field support. Barbara Bridges, Joan Dean, Graeme Taylor and Andrew Swan are thanked for their contributions to the laboratory measuring program.

References

Abrahamsen, N., 1985. Possible types of rotations and translations in the Scandinavian Caledonides. J. Geodyn., 2: 245–263.

Abrahamsen, N., Wilson, J.R., Thy, P., Olsen, N.O. and Esbensen, K.H., 1979. Palaeomagnetism of the Fongen–Hyllingen gabbro complex, southern Scandinavian Caledonides; plate rotation or polar shift? Geophys. J.R. Astron. Soc., 59: 231–248.

Andressen, A., Fareth, E., Bergh, S., Kristensen, S.E. and Krogh, E., 1985. Review of Caledonian lithotectonic units in Troms, North Norway. In: D.G. Gee and B.A. Sturt (Editors), The Caledonide Orogen–Scandinavia and Related Areas. Wiley, Chichester, pp. 569–578.

Barnes, S.J., 1990. Sulphide segregation history of the Tverrfjell portion of the Råna Layered Intrusion, Norway. IMM Special Issue for the 5th Magmatic Sulphide Conference, Zimbabwe, 1987, in press.

Boyd, R. and Mathieson, C.O., 1979. The nickel mineralisation of the Råna mafic intrusion, Nordland, Norway. Can. Mineral., 17: 287–298.

Boyle, A.P., 1980. The Sulitjelma amphibolites, Norway: part of a Lower Palaeozoic ophiolite complex? In: A. Panayiotou (Editor), Proceedings of the International Ophiolite Symposium, Cyprus, 1979. Cyprus Geol. Surv. Dep., Nicosia, pp. 567–575.

Boyle, A.P., Griffiths, A.J. and Mason, R., 1979. Stratigraphical inversion in the Sulitjelma area, Central Scandinavian Caledonides. Geol. Mag., 116: 393–402.

Briden, J.C., Morris, W.A. and Piper, J.D.A., 1973. Palaeomagnetic studies in the British Caledonides. VI. Regional and global implications. Geophys. J.R. Astron. Soc., 34: 107–134.

Briden, J.C., Turnell, H.B. and Watts, D.R., 1984. British palaeomagnetism, Iapetus Ocean and the Great Glen Fault. Geology, 12: 428–431.

Bylund, G. and Zellman, O., 1980. Palaeomagnetism of the dolerites of the Sarv Nappe, southern Swedish Caledonides. Geol. Foren. Stockholm Forh., 102: 393–402.

Chamalaun, F.H. and Creer, K.M., 1964. Thermal demagnetisation studies of the Old Red Sandstone of the Anglo-Welsh cuvette. J. Geophys. Res., 69: 1607–1616.

Claesson, S., 1979. Early Palaeozoic geomagnetism of Gotland. Geol. Foren. Stockholm Forh., 101: 149–155.

Creer, K.M., 1968. Palaeozoic palaeomagnetism. Nature, 219: 246–250.

Douglass, D.N., 1988. Palaeomagnetics of Ringerike Old Red Sandstone and related rocks, Southern Norway; implications for pre-Carboniferous separation of Baltica and British Terranes. Tectonophysics, 148: 11–27.

Esang, C.B. and Piper, J.D.A., 1984. Palaeomagnetism of Caledonian intrusive suites in the Northern Highlands of Scotland: constraints to tectonic movements within the Caledonian orogenic belt. Tectonophysics, 104: 1–34.

Foslie, S., 1921. Field observations in northern Norway bearing on magmatic differentiation. J. Geol., 29: 701–719.

Gee, D.G., Ghezou, J-C., Roberts, D. and Wolff, F.C., 1985. The central-southern part of the Scandinavian Caledonides. In: D.G. Gee and B.A. Sturt (Editors), The Caledonide Orogen—Scandinavia and Related Areas. Wiley, Chichester, pp. 109–134.

Gustavson, M., 1969. The Caledonian mountain chain of the southern Troms and Ofoten Areas. Nor. Geol. Unders., 261: 110 pp.

Hossack, J.R. and Cooper, M.A., 1986. Collision tectonics in the Scandinavian Caledonides. In: M.A. Coward and A.C. Ries (Editors), Collision Tectonics. Geol. Soc. London, Spec. Publ., 19: 287–304.

Irving, E. and Strong, D.F., 1985. Palaeomagnetism of rocks from the Burin Peninsula, Newfoundland: hypothesis of late Palaeozoic displacement of Acadia criticised. J. Geophys. Res., 90: 1949–1962.

Kautsky, G.A., 1953. Der geologische Bau des Sulitjelma–Salojauregebietes in den nord-Skandinavischen Kaledoniden. Sver. Geol. Unders., C528: 233 pp.

Kent, D.V. and Opdyke, N.D., 1985. Multicomponent magnetisations from the Mississippian Mauch Chunk Formation of the central Appalachians and their tectonic implications. J. Geophys. Res., 90: 5371–5383.

Lie, L.G., Storetvedt, K.M. and Gjellestad, G., 1969. The palaeomagnetism of the Kvamshesten Old Red sequence, southwest Norway. Nord. Geol. Tidsskr., 49: 241–249.

Løvlie, R. and Mitchell, J.G., 1982. Complete remagnetisation of some Permian dykes from western Norway induced during burial/uplift. Phys. Earth Planet. Inter., 30: 415–421.

MacDonald, W.D., 1980. Net tectonic rotation, apparent tectonic rotation and the structural tilt correction in palaeomagnetic studies. J. Geophys. Res., 85: 3649–3669.

Mason, R., 1971. The chemistry and structure of the Sulitjelma Gabbro. Nor. Geol. Unders., 269: 108–142.

Mason, R., 1980. Temperature and pressure estimates in the contact aureole of the Sulitjelma gabbro: implications for an ophiolite origin. In: A. Panayiotou (Editor), Proceedings of the International Ophiolite Symposium, Cyprus, 1979. Cyprus Geol. Surv. Dep., Nicosia, pp. 576–581.

McClelland Brown, E., 1983. Palaeomagnetic studies of fold development and propogation in the Pembrokeshire Old Red Sandstone. Tectonophysics, 98: 131–149.

Miller, J.D. and Kent, D.V., 1986. Syn folding and prefolding magnetisations in the Upper Devonian Catskill Formation of eastern Pennsylvania. J. Geophys. Res., 91: 12791–12803.

Minsaas, O. and Sturt, B.A., 1985. The Ordovician–Silurian clastic sequence overlying the Lyngen Gabbro Complex and its environmental significance. In: The Caledonide Orogen—Scandinavia and Related Areas. D.G. Gee and B.A. Sturt (Editors), Wiley, Chichester, pp. 379–394.

Mulder, F.G., 1965. Palaeomagnetic investigation in the Vanern district (Sweden). Geol. Mijnbouw, 44: 307–312.

Perroud, H. and Van der Voo, R., 1985. Palaeomagnetism of the late Ordovician Thouars Massif, Vendee Province, France. J. Geophys. Res., 90: 4611–4625.

Phillips, W.E.A., Stillman, C. and Murphy, T., 1976. A Caledonian plate tectonic model. J. Geol. Soc. London, 132: 579–609.

Piper, J.D.A., 1974. Sulitjelma gabbro, northern Norway: a palaeomagnetic result. Earth Planet. Sci. Lett., 21: 383–388.

Piper, J.D.A., 1987. Palaeomagnetism and the Continental Crust. Open University Press, Milton Keynes, 434 pp.

Piper, J.D.A. and Poppleton, T.J., 1990. A palaeomagnetic study of the Basal Gneiss Region and Jotun Nappe complex, central-southern Scandinavian Caledonides. Geophys. J., Int., 101: 591–612.

Poppleton, T.J. and Piper, J.D.A., 1990. A palaeomagnetic study of a Caledonian ophiolite Karmoy, SW Norway. Phys. Earth Planet. Inter., 61: 297–314.

Randall, B.A.O., 1971a. The igneous rocks of the Lyngen Peninsula, Troms, Norway. Nor. Geol. Unders., 269: 143–146.

Randall, B.A.O., 1971b. An outline of the geology of the Lyngen Peninsula, Troms, Norway. Nor. Geol. Unders., 269: 68–71.

Roberts, D.G. and Gale, G.H., 1978. The Caledonian–Appalachian Iapetus Ocean. In: D.H. Tarling (Editor), Evolution of the Earth's crust. Academic Press, London, pp. 255–342.

Roberts, D.G. and Gee, D.G., 1985. An introduction to the structure of the Scandinavian Caledonides. In: D.G. Gee and B.A. Sturt (Editors), The Caledonian Orogen—Scandinavia and Related Areas. Wiley, Chichester, pp. 55–68.

Roy, J.L. and Morris, W.A., 1983. A review of palaeomagnetic results from the Carboniferous of North America; the concept of Carboniferous geomagnetic field horizon markers. Earth Planet. Sci. Lett., 65: 167–181.

Sallomy, J.T. and Piper, J.D.A., 1973a. Palaeomagnetic studies in the British Caledonides. IV. Lower Devonian lavas of the Strathmore region Scotland. Geophys. J.R. Astron. Soc., 34: 49–64.

Sallomy, J.T. and Piper, J.D.A., 1973b. Palaeomagnetic studies in the British Caledonides. II. The Younger Gabbros of Aberdeenshire, Scotland. Geophys. J.R. Astron. Soc., 34: 13–26.

Stephens, M.B., Gustavson, M., Ramberg, I.B. and Zachrisson, R., 1985a. The Caledonides of central-northern Scandinavia—a tectonostratigraphic overview. In: D.G. Gee and B.A. Sturt (Editors), The Caledonide Orogen—Scandinavia and Related Areas. Wiley, Chichester, pp. 135–162.

Stephens, M.B., Furnes, H., Robins, B. and Sturt, B.A., 1985b. Igneous activity within the Scandinavian Caledonides. In: D.B. Gee and B.A. Sturt (Editors), The Caledonide Orogen—Scandinavia and Related Areas. Wiley, Chichester, pp. 623–656.

Storetvedt, K.M. and Gjellestad, G., 1966. Palaeomagnetic investigation of an Old Red Sandstone formation of southern Norway. Nature (London), 212: 59–61.

Storetvedt, K.M., Halvorsen, E. and Gjellestad, G., 1978. Thermal analysis of the natural remanent magnetism of some Upper Silurian red sandstones in the Oslo region. Tectonophysics, 5: 413–426.

Taylor, G.K., 1988. A palaeomagnetic study of a Caledonian ophiolite. Geophys. J., 94: 157–166.

Torsvik, T.H., 1985a. Magnetic properties of the Lower Old Red Sandstone lavas in the Midland Valley, Scotland; palaeomagnetic and tectonic considerations. Phys. Earth Planet. Inter., 39, 194–207.

Torsvik, T.H., 1985b. Palaeomagnetic results from the Peterhead granite, Scotland: implications for regional late Caledonian magnetic overprinting. Phys. Earth Planet. Inter., 39: 108–117.

Torsvik, T.H., Løvlie, R. and Storetvedt, K.M., 1983. Multicomponent magnetisation in the Helmsdale granite, North Scotland; geotectonic implications. Tectonophysics, 98: 111–129.

Torsvik, T.H., Sturt, B.A., Ramsay, D.M., Kisch, H.J. and Bering, D., 1986. The tectonic implications of Solundian (Upper Devonian) magnetisation of the Devonian rocks of Kvamshesten, western Norway. Earth Planet. Sci. Lett., 80: 337–347.

Torsvik, T.H., Sturt, B.A., Ramsay, D.M. and Vetti, V., 1987. The tectono-magnetic signature of the Old Red Sandstone and Pre-Devonian strata in the Hasteinen area, western Norway, and implications for later stages of the Caledonian orogeny. Tectonics, 6(3): 305–322.

Torsvik, T.H., Sturt, B.A., Ramsay, D.M., Bering, D. and Fluge, P.R., 1988. Palaeomagnetism, magnetic fabrics and the structural style of the Hornalon Old Red Sandstone, western Norway. J. Geol. Soc. London, 145: 413–430.

Turner, P., 1977. Remanent magnetism of Middle Old Red Sandstone lacustrine and fluviatile sediments from the Orcadian basin, Scotland. J. Geol. Soc. London, 133: 37–50.

Turner, P. and Tarling D.H., 1975. Implications of new palaeomagnetic results from the Carboniferous System of Britain, J. Geol. Soc. London, 131: 469–488.

Turner, P., Vaughan, D.J. and Tarling, D.H., 1978. Palaeomagnetic and mineralogical studies of Devonian lacustrine sediments from Caithness, Scotland. Phys. Earth Planet. Inter., 16: 73–83.

Watts, D.R., 1982. A multicomponent dual polarity palaeomagnetic regional overprint from the Moine of North-west Scotland. Earth Planet. Sci. Lett., 61: 190–198.

Watts, D.R., 1985. Palaeomagnetic resetting in the Barrovian zones of Scotland and its relationship to the later structural history. Earth Planet. Sci. Lett., 75: 258–264.

Watts, D.R. and Briden, J.C., 1984. Palaeomagnetic signature of slow post-orogenic cooling of the north-west Highlands of Scotland recorded in the Newer Gabbros of Aberdeenshire. Geophys. J.R. Astron. Soc., 77: 775–778.

Wilson, M.R., 1981. Geochronological results from Sulitjelma, Norway. Terra Cognita, 1: 82.

Wilson, R.L. and Everitt, C.W.F., 1963. Thermal demagnetisation of some Carboniferous lavas for palaeomagnetic purposes. Geophys. J.R. Astron. Soc., 8: 149–164.